INCH-POUND

MIL-PRF-83870B (USAF)
AMENDMENT 1
26 March 1999

PERFORMANCE SPECIFICATION

FILTER ELEMENT, DISPOSABLE, 20 MICRON ABSOLUTE,

CONSTANT SPEED DRIVE, AIRCRAFT

This amendment forms a part of MIL-PRF-83870B (USAF) dated , 31 December 1998 and is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Pages

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PAGE 4

3.7 a.: Add:

"(Optional)" after Manufacturer's name

PAGE 8

4.3.2: Delete and substitute:

"Sampling. A sample of filter elements shall be selected in accordance with ANSI/ASQ Z1.4 at an S-1 inspection level unless modified in the procurement document (see 6.2), and shall be subjected to the pressure build-up and collapse test in 4.6.6, pressure drop (clean element) part only, and the pressure drop (clean element) shall not exceed the value specified in table II."

Table IV: Delete and substitute:

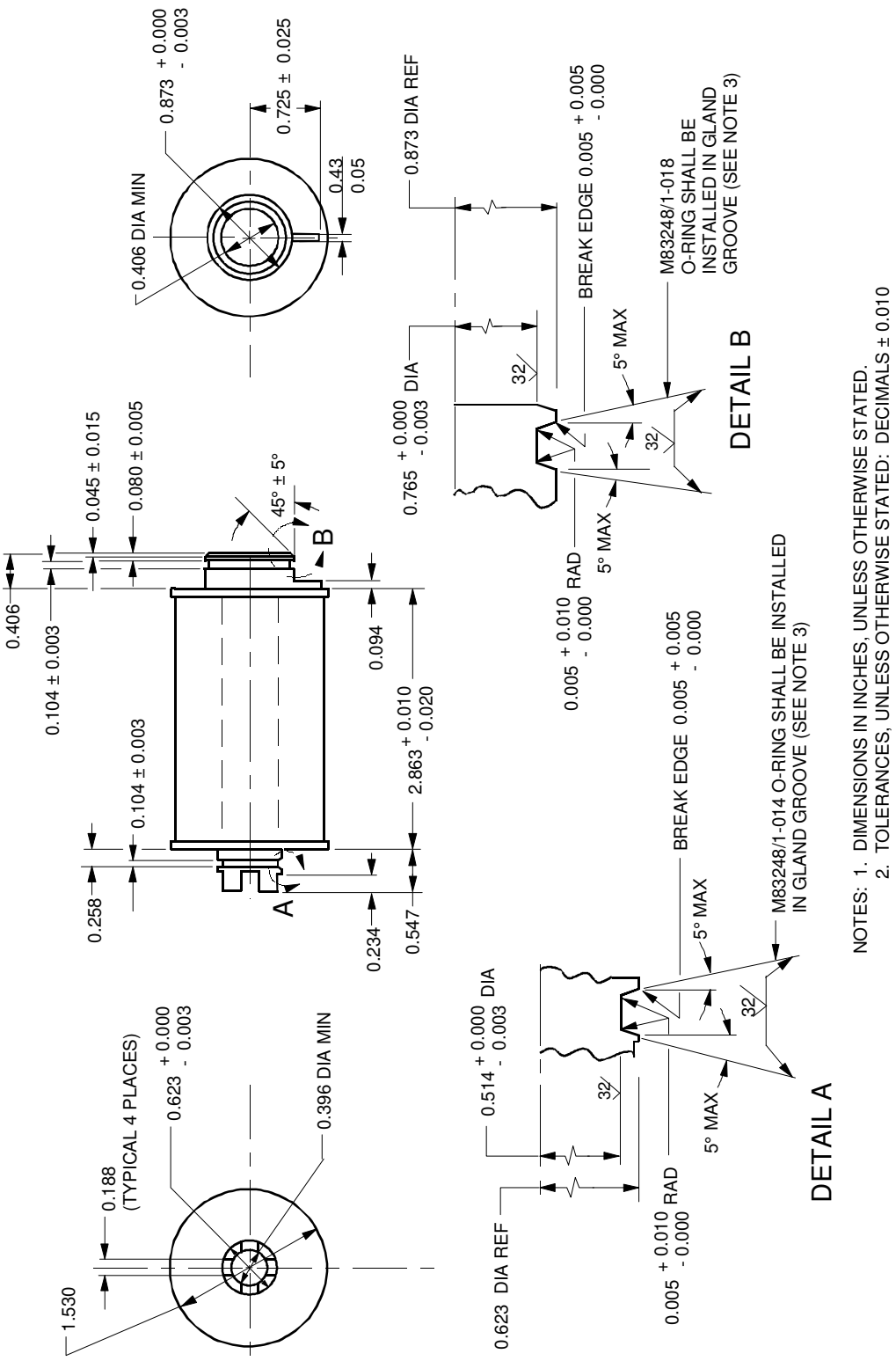
In the last column under Verification, second row, change "4.6.4" to "4.6.8".

Custodians:
Air Force - 99

Preparing activity:
Air Force - 71

(Project 1650-F187)

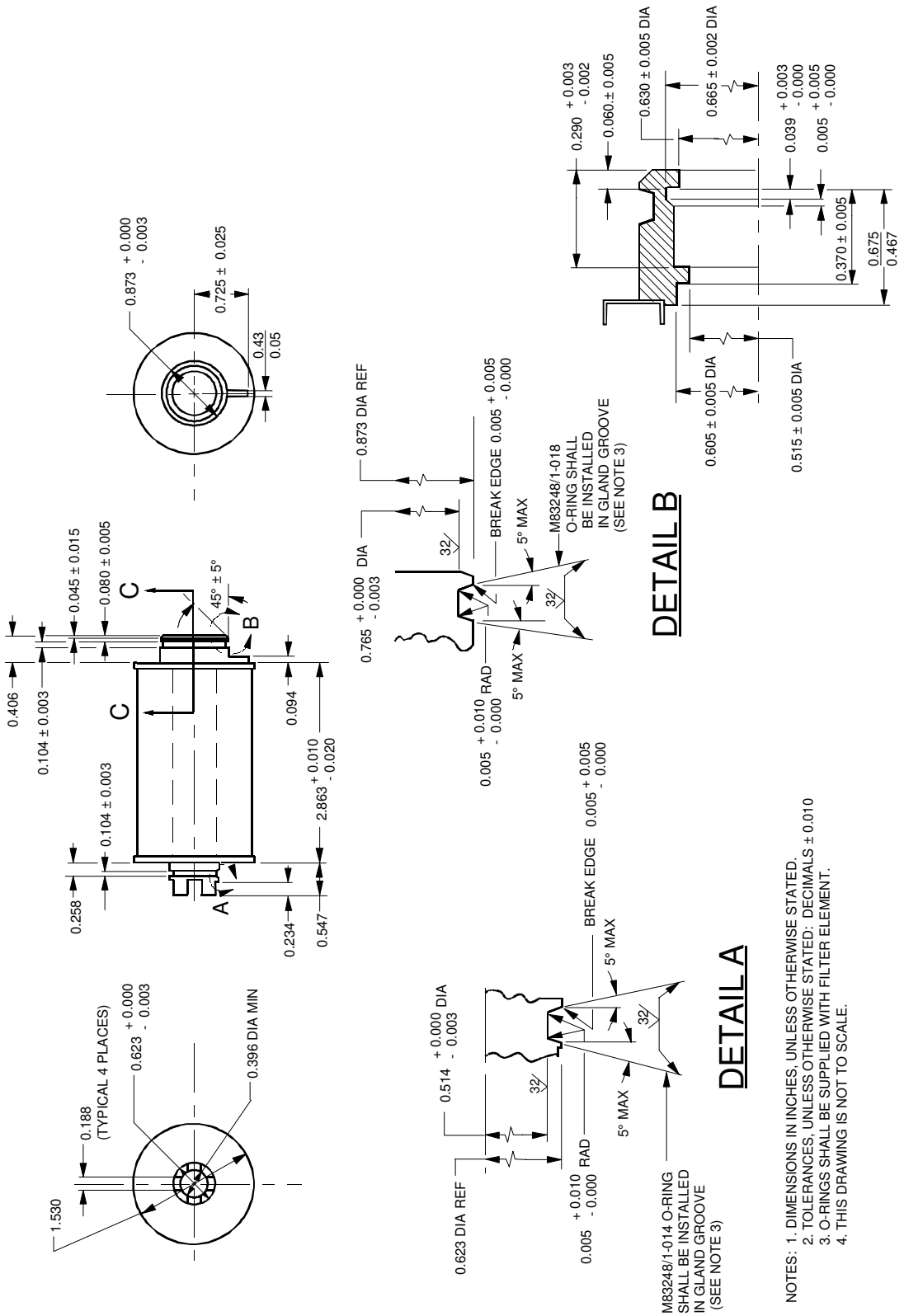
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NOTES: 1. DIMENSIONS IN INCHES, UNLESS OTHERWISE STATED.
2. TOLERANCES, UNLESS OTHERWISE STATED: DECIMALS ± 0.010
3. O-RINGS SHALL BE SUPPLIED WITH FILTER ELEMENT.
4. THIS DRAWING IS NOT TO SCALE.

FIGURE 3. M83870-4 filter element dimensions

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4. THIS DRAWING IS NOT TO SCALE.

FIGURE 4. M83870-5 filter element dimensions

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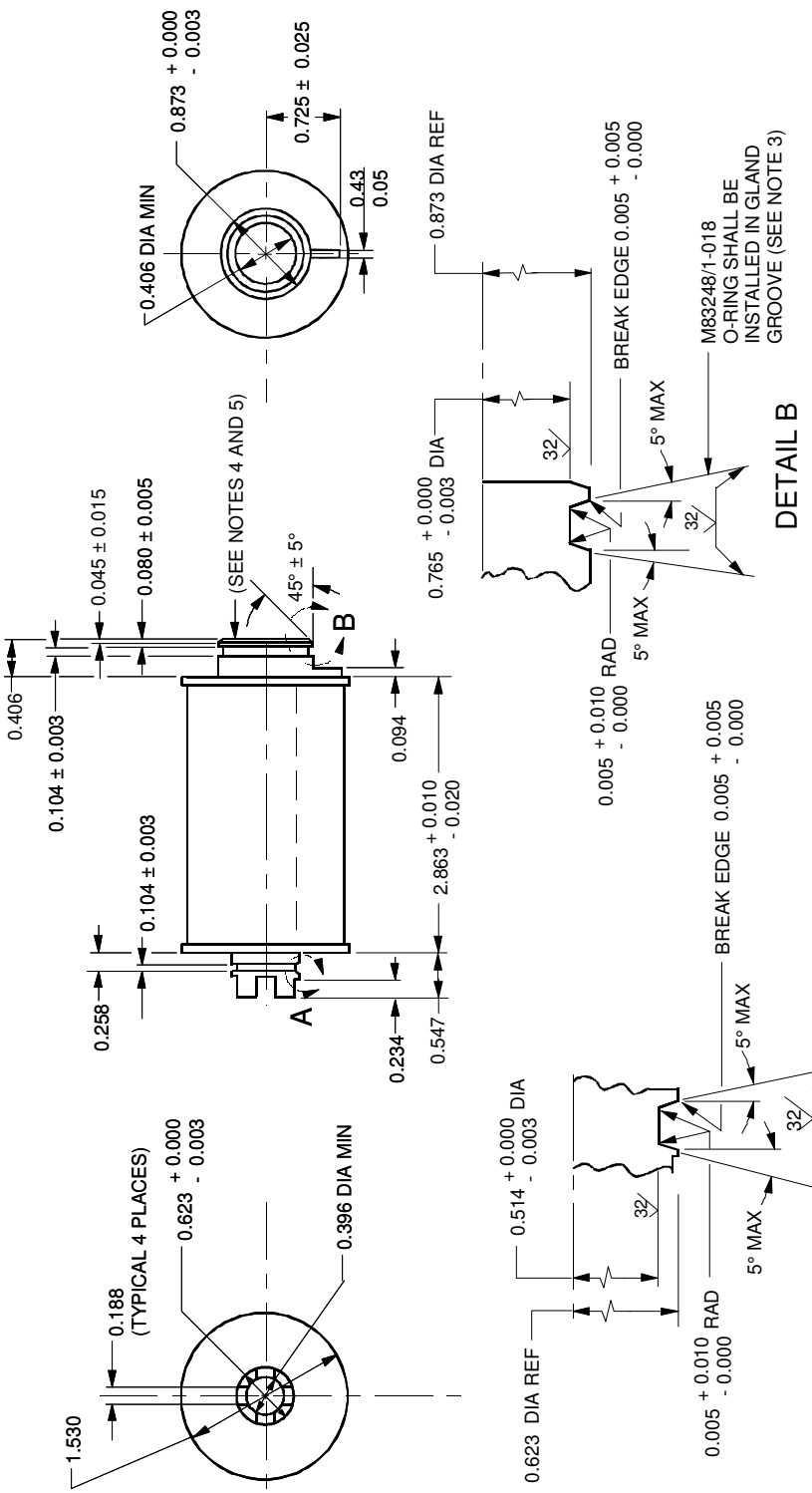
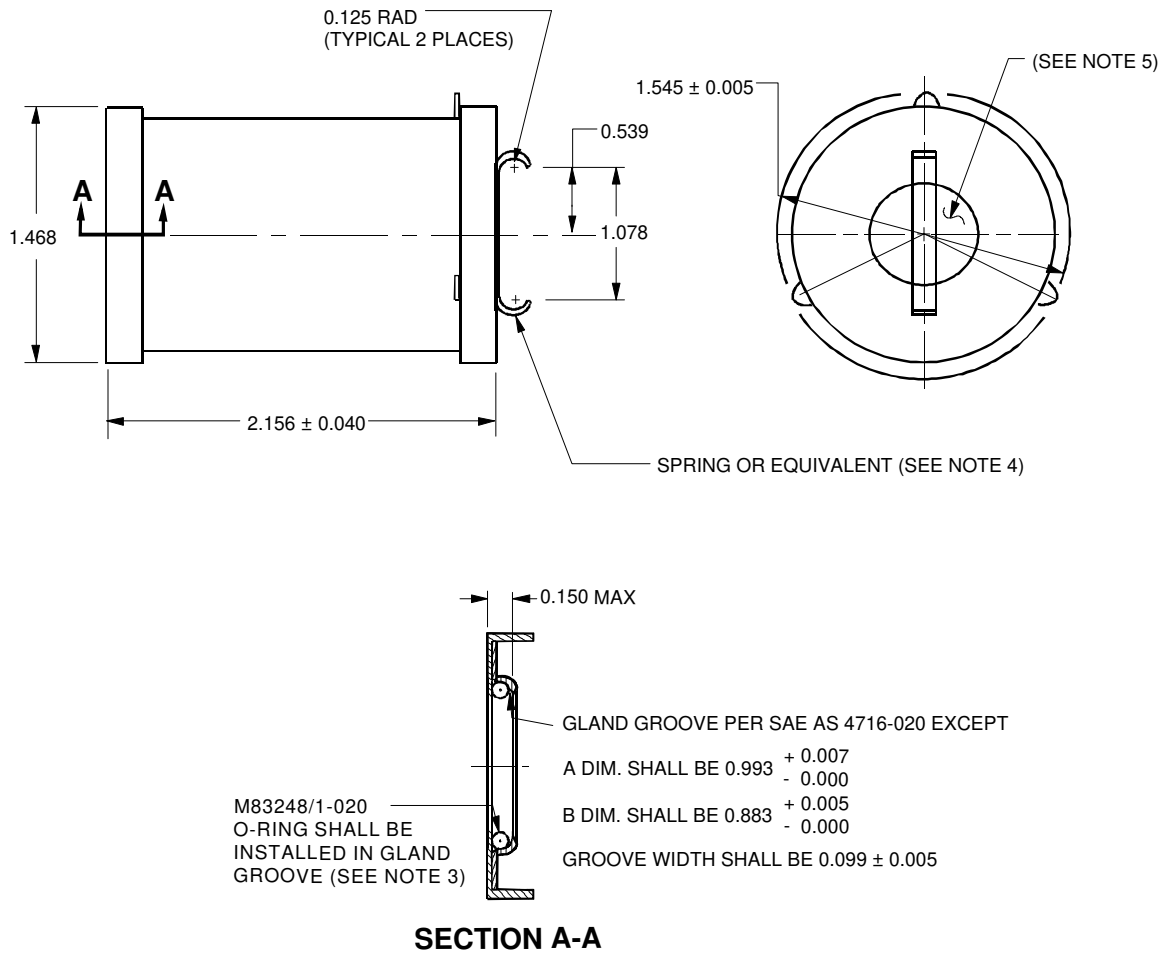


FIGURE 5. M83870-6 filter element dimensions

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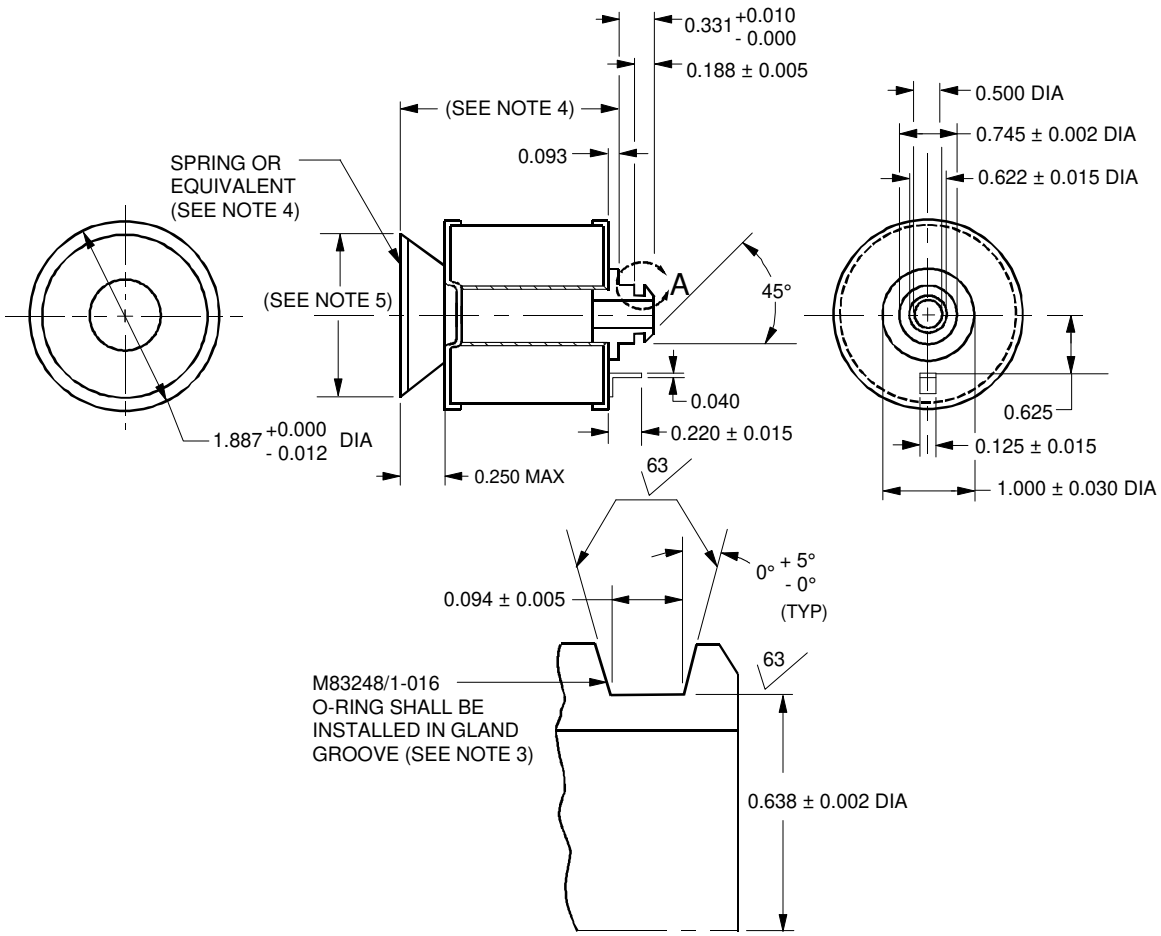


SECTION A-A

NOTES: 1. DIMENSIONS IN INCHES, UNLESS OTHERWISE STATED.
2. TOLERANCES, UNLESS OTHERWISE STATED: DECIMALS ± 0.010.
3. O-RING SHALL BE SUPPLIED WITH FILTER ELEMENT.
4. SPRING CHARACTERISTICS MUST BE SUCH AS TO PROVIDE AXIAL LOADING OF 5 TO 15 POUNDS WHEN COMPRESSED 0.09 INCH.
5. RELIEF VALVE THIS END.
6. THIS DRAWING IS NOT TO SCALE.

FIGURE 6. M83870-7 filter element dimensions

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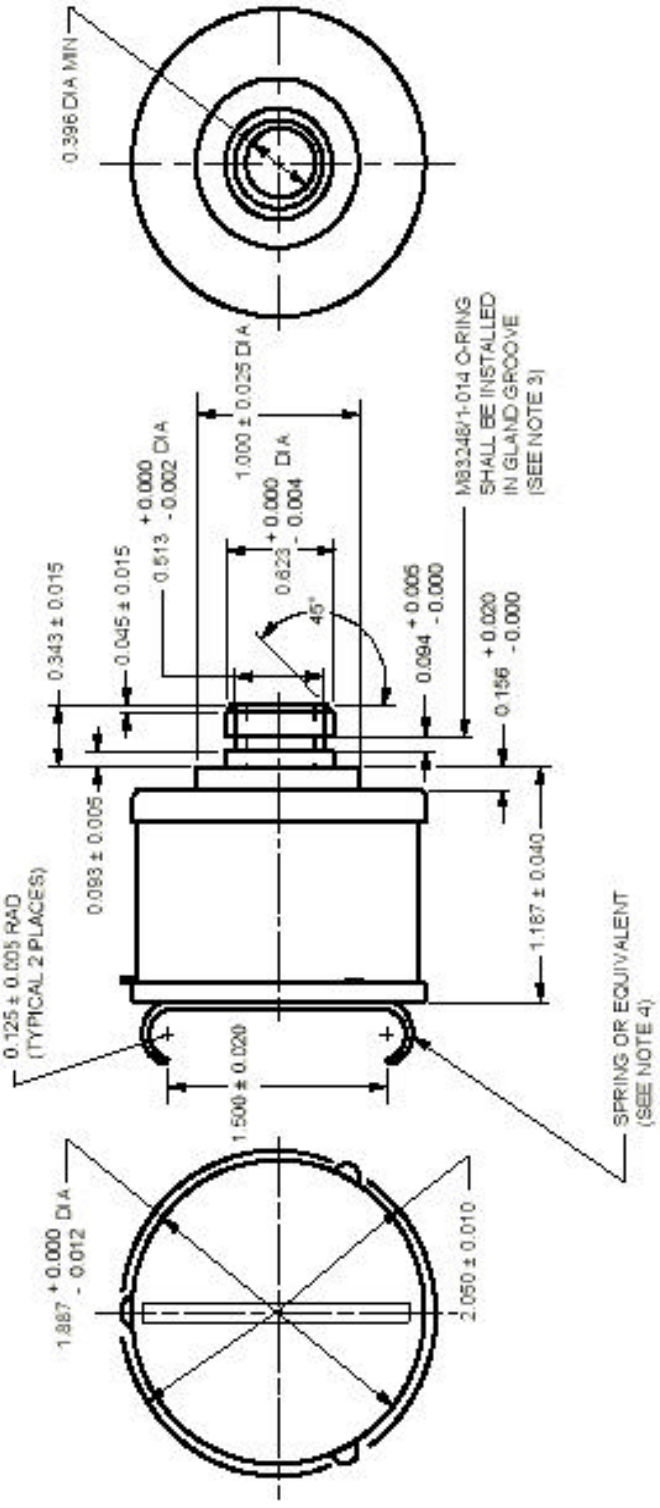


DETAIL A

NOTES: 1. DIMENSIONS IN INCHES, UNLESS OTHERWISE STATED.
2. TOLERANCES: DECIMALS ± 0.010
ANGLES ± 3°
3. O-RING SHALL BE SUPPLIED WITH FILTER ELEMENT.
4. ELEMENT LENGTH AND SPRING CHARACTERISTICS MUST BE SUCH AS TO PROVIDE AXIAL LOADING OF 10 TO 50 POUNDS WHEN COMPRESSED TO 1.670 ± 0.010 INCH.
5. MAXIMUM DIA WHEN COMPRESSED TO 1.670 ± 0.010 INCH IS 1.75 INCH. MINIMUM DIA WHICH WILL CONTACT FILTER BOWL IS 0.812 INCH.
6. THIS DRAWING IS NOT TO SCALE.

FIGURE 7. M83870-8 filter element dimensions

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NOTES: 1. DIMENSIONS IN INCHES, UNLESS OTHERWISE STATED.
2. TOLERANCE: ANGLE ± 3°.
3. O-RING SHALL BE SUPPLIED WITH FILTER ELEMENT.
4. SPRING CHARACTERISTICS MUST BE SUCH AS TO PROVIDE AXIAL LOADING OF 5 TO 15 POUNDS WHEN COMPRESSED 0.09 INCH.

FIGURE 8. M83870-9 filter element dimensions

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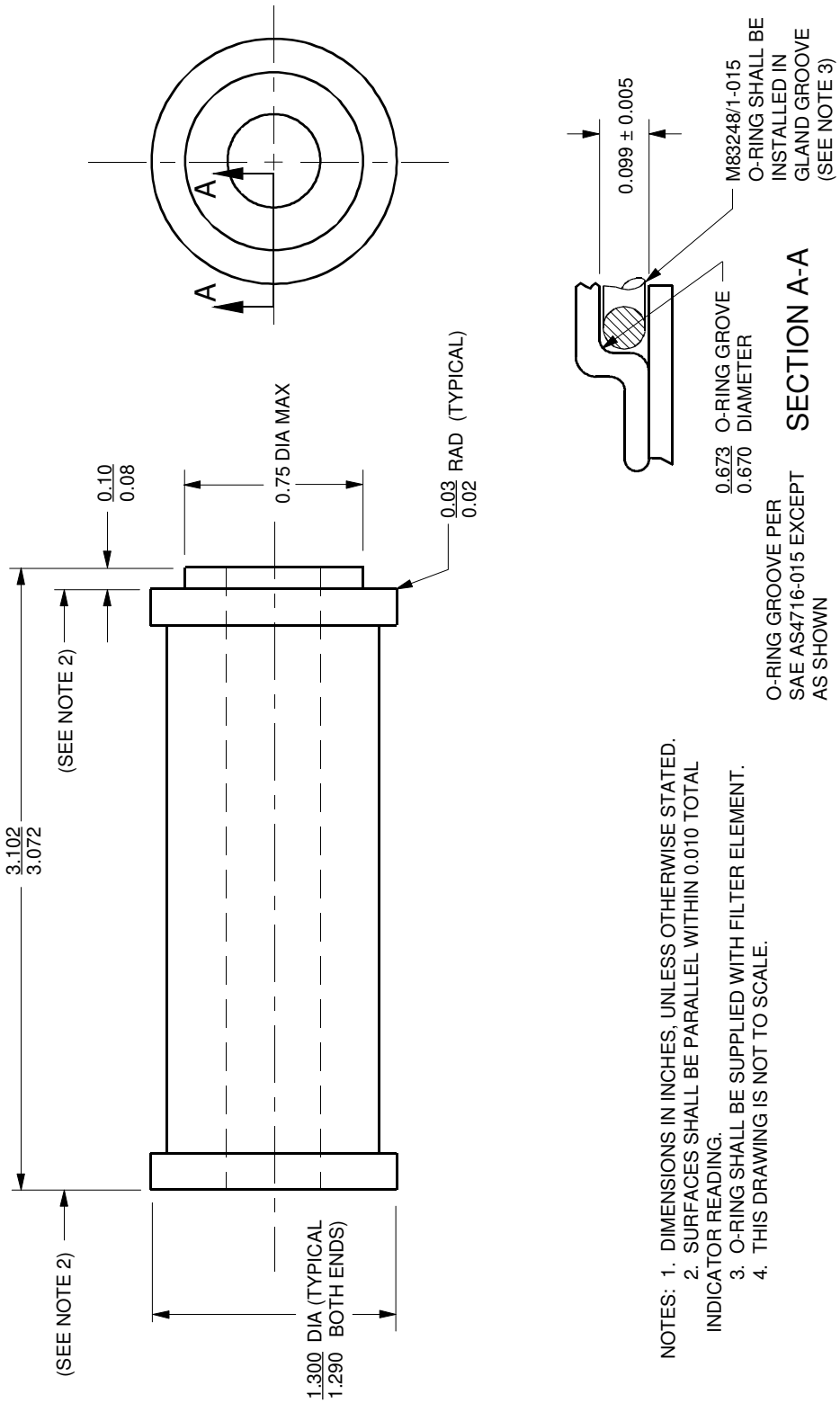


FIGURE 9. M83870-10 filter element dimensions

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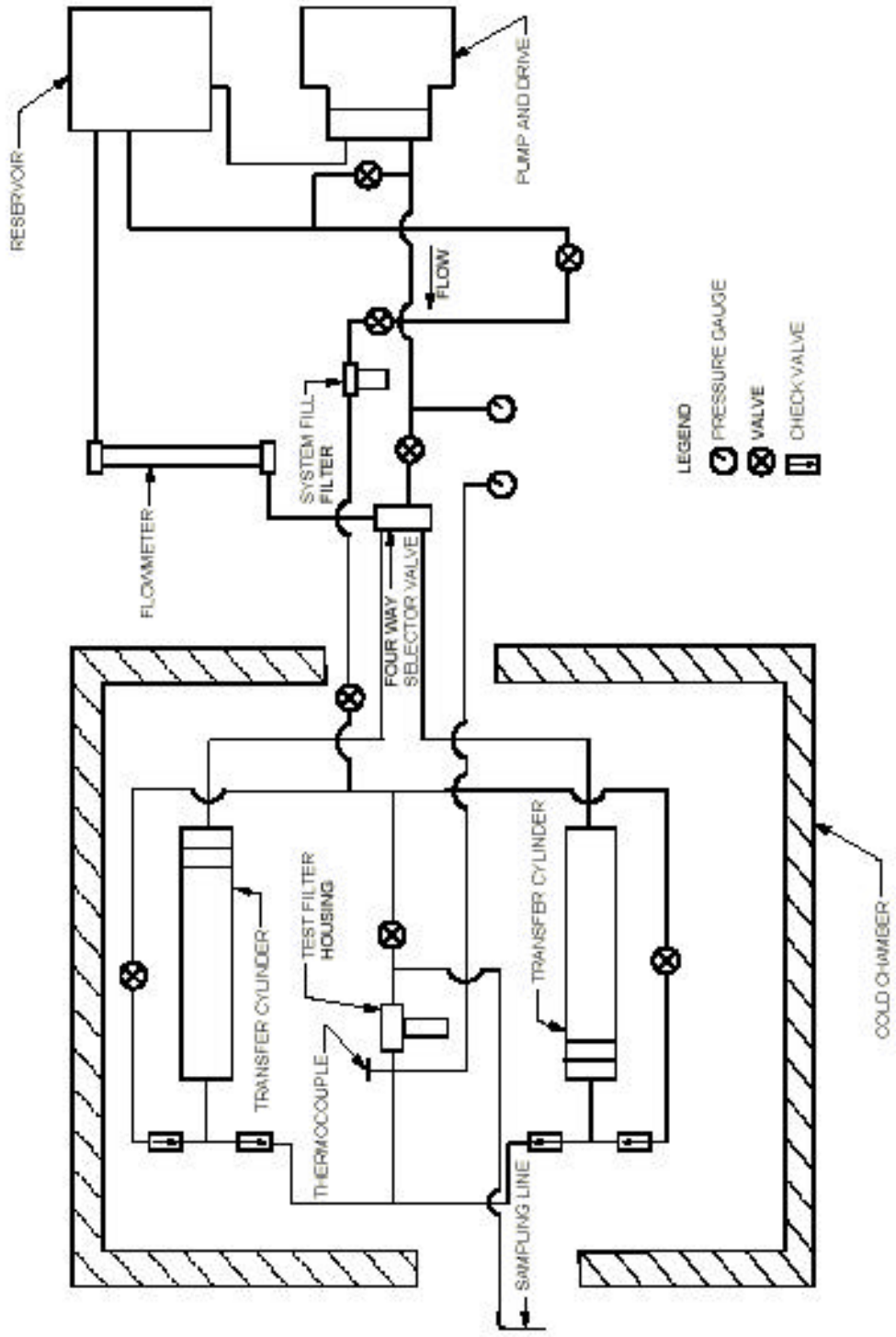


FIGURE 10. Typical setup for cold start test

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